
Dollars, Debt, and Destiny: Marfrig's Dilemma in a Changing Currency World

Teaching Note

Synopsis

Marfrig Global Foods (Marfrig) CFO Tang David must decide whether to anchor new short-term and long-term contracts in US dollars. For decades, the dollar had functioned as the global safe asset, but David wondered whether headwinds on the US policy front threatened the dollar's dominance. The case offers historical context about the dollar's rise to prominence and examines the dollar's current status in relation to US tariff policy, US fiscal stability, and firm-level outcomes.

This case is the second piece in a two-class series that covers tariffs and the US dollar.¹ With the 2025 US tariffs as a backdrop, the case covers the evolving role of the US dollar as an international safe asset. Students will not only understand why the dollar holds a preeminent role in international trade and finance, but they will learn how certain shocks can imperil the dollar's position and influence global business transactions.

Learning Objectives

1. Understand why global firms transact in US dollars and how this choice exposes them to macroeconomic risk.
2. Use the Three-Paned Model to explain how capital flows and policy uncertainty affect exchange rates.
3. Analyze fiscal sustainability using a present discounted value framework and explore how rising interest rates affect debt dynamics.
4. Contrast model predictions with observed market behavior to identify shifts in investor confidence.

¹ For the companion case, see Felipe Saffie and Bryan Harrison, "Tariff Trouble: Navigating a Trade War in the Global Supply Chain," UVA-GEM-0255 (Charlottesville, VA: Darden Business Publishing, 2026).

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Proposed assignment

Reading

- “Dollars, Debt, and Destiny: Marfrig’s Dilemma in a Changing Currency World” (UVA-GEM-0254)²

Questions for advance assignment

Read the case and answer the following questions.

1. Why does Marfrig typically invoice export contracts and borrow in US dollars? What advantages does the dollar offer to a multinational firm, and what risks emerge when the dollar weakens?
2. According to the Three-Paned Model, what should happen to the value of the dollar following a broad US tariff? Do the data in case Exhibits 5a through 5d support this prediction? How do you explain the data?
3. Rising US interest rates increase debt service costs and raise questions about fiscal sustainability. Using the present discounted value identity, explain how higher rates could reinforce pressure on the dollar.
4. Marfrig is finalizing a three-year export contract and a 30-year bond issuance. Given the recent volatility, should both deals be denominated in US dollars? Why or why not?

Teaching Plan

The instructor can use this case to help students understand the aspects of US dollar supremacy and the vulnerability of dollar dominance. This case was written for use in the second half of the “Global Economies and Markets” (GEM) core course at the University of Virginia Darden School of Business. The first half of the course teaches the IS/LM and AS/AD models in closed-economy settings, while the second half of the course uses the Three-Paned Model, an open-economy extension of the IS/LM model.³ The plan assumes that students have knowledge of these macroeconomic concepts as well as an understanding of tariffs and US tariff policy (covered in the companion case, UVA-GEM-0255). Prior study of dominant currencies or foreign exchange markets is not required, but familiarity with US fiscal policy could be helpful.

The following plan is for an 85-minute class session. We break the class discussion into several segments. This teaching note describes the essential aspects of each segment in detail, and at the end of each segment discussion, we include a bullet-point summary of one approach to structuring the discussion. In the summary, we italicize questions that could be posed to the class.

After an opening discussion about the role of the US dollar in business transactions, the case largely focuses on dollar dynamics related to the Triffin dilemma, the US tariff shock in 2025, and the concept of US debt sustainability. In the end, students will apply their understanding of these forces to determine how Marfrig should anchor new contracts.

² Felipe Saffie and Bryan Harrison, “Dollars, Debt, and Destiny: Marfrig’s Dilemma in a Changing Currency World,” UVA-GEM-0254 (Charlottesville, VA: Darden Business Publishing, 2026).

³ AS = aggregate supply; AD = aggregate demand; IS = investment–saving; LM = liquidity preference–money supply.

Segment 0: Motivation (Why the US dollar?) (10 min.)

To stimulate discussion about the key issues of the case, the instructor can begin by asking, “What is the case about?” The instructor should make sure students mention some of the key case details, including the decision about whether the firm should anchor new short- and long-term deals in US dollars. When contextualizing the decision, students need to recognize a central detail: Marfrig is a non-US business that conducts many transactions in dollars, even though most of its sales come from outside of the United States (case Exhibit 8a).⁴ After students highlight this point, the instructor can follow up by asking, “Why does Marfrig (or any non-US firm) use the dollar?” This allows for a general dialogue about the role of the dollar in international trade and finance. The dollar serves as the global reserve currency, and many international firms conduct transactions in dollars, creating global network externalities. For businesses across the world, the dollar represents a safe, stable, and liquid asset, making transactions easier and less costly. At the individual firm level, invoicing and issuing debt in dollars provides a natural hedge; as the value of the dollar changes, one side of the balance sheet might be hurt, but the other side of the balance sheet might benefit.⁵ While the class discusses the advantages of dollar utilization, the instructor can list some of the student responses on the board (see **Board plan 1**).

The following questions can facilitate the discussion for this segment:

- *What is the case about?* Marfrig, a Brazilian company that conducts many transactions in US dollars, is trying to determine how to anchor new short-term and long-term contracts.
- *Why does Marfrig (or any non-US firm) use the dollar?* See **Board plan 1**.

Segment 1: Supply and demand of the US dollar (safe-asset view) (10 min.)

The initial discussion about dollar usage can lead to dialogue about supply and demand of the dollar. To start this portion of class, the instructor can ask a general question: “What determines the price of a product?” When students identify supply and demand as the factors that determine price, the instructor can begin to model the foreign exchange market for dollar-denominated assets. As the instructor illustrates the model, the class can discuss specific factors that affect the supply and demand for the dollar (see **Board plan 1**). Then, after showing an outward shift of the demand curve, the instructor can ask, “What does it mean if the dollar is in high demand?” Students should recognize that the dollar would appreciate, which means the currency becomes more expensive—therefore, foreigners might not want to buy as many US products, and US exporters lose competitiveness. The instructor can follow up by asking, “How can the United States counter the loss of competitiveness?” Students should see that supplying dollars (shifting the supply curve outward) lowers the exchange rate; of particular relevance to the case, issuing debt (supplying US Treasuries) can increase the amount of dollars on the foreign exchange market over time. This implies that the country providing the world’s safe asset has the option to accept the consequences of the overvalued currency or supply debt.

The following questions can facilitate the discussion for this segment:

- *What determines the price of a product?*
 - Supply and demand

⁴ Case Exhibit 3 shows international debt data, and case Exhibit 4 shows global invoicing and exporting data.

⁵ Businesses naturally hold some assets and liabilities in the same foreign currency. This means that when the exchange rate shifts, changes on the asset side of the balance sheet are offset by changes on the liabilities side of the balance sheet.

- *What does it mean if the dollar is in high demand?*
 - The dollar appreciates (becomes more expensive)
 - US exporters lose competitiveness
- *How can the United States counter the loss of competitiveness?*
 - Supply dollars (issue debt)

Segment 2: The Triffin dilemma (modernized) (20 min.)

To start this segment, students can think further about issues related to government debt. The instructor can begin by asking, “What determines the value of a US Treasury?” Students should understand that people view US government debt as a safe asset because of the government’s ability to pay. To meet its debt obligations, the government has the facility to refinance—to pay off existing debt through the issuance of new debt. After this brief discussion about the dynamics of US government debt, the instructor can ask, “What makes a country default on its debt?” Students should recognize that a government risks default when it can no longer refinance its debt, which can cause interest rates to rise as people lose confidence in the government and ultimately stop lending. The instructor should make sure students realize that interest rates should be closely watched because higher rates can place additional burden on the government’s ability to refinance and pay its obligations (when interest rates rise, it becomes more costly to service the debt).

Students should also consider how an economy’s growth relates to its ability to pay off debt. To begin this portion of the discussion, the instructor can ask, “How does growth in gross domestic product (GDP) affect a country’s ability to repay debt?” Students should understand that the ability to pay off debt grows as the economy grows (the tax base becomes larger). If developing countries grow faster than the United States, demand for the dollar will increase, and the United States will need to supply additional debt. This implies that the United States must provide a safe asset at a pace faster than GDP growth, which can impede its ability to repay debt.

The initial class dialogue underscores the fundamental concepts of the modernized Triffin dilemma. After the preliminary discussion about dominant currencies and debt sustainability, the instructor can ask, “What can we associate with the reasons that dominant currencies fall?” Once students identify the Triffin dilemma (case Exhibit 6) and describe the paradox, the instructor can chart it on the board (see **Board plan 2**). To relate the Triffin dilemma to the current circumstances of the United States, the instructor can ask, “Will the dollar remain the dominant currency in 5 years? 10 years? 200 years?” Students might say that the dollar should continue to reign supreme in the immediate future but likely will not dominate a century from now. At some point, due to massive government debt or the rise of more viable alternatives (see case Exhibits 2a, 2b, 5b and 5c for data on dollar alternatives), the safety of the dollar will be questioned, but the timing of the dollar’s fall is uncertain.⁶ While students might want to discuss the future role of the dollar further, the instructor can postpone more significant discussion until Segment 5, when students consider the prospects of the dollar relative to the firm’s decision.

The following questions can facilitate discussion for this segment:

- *What determines the value of a US Treasury?*
 - The government’s ability to pay (refinance the debt)

⁶ History suggests that dominant currencies reign for a limited period (case Exhibit 1), but the instructor should ensure that students understand why these currencies can eventually lose their status.

- *What makes a country default on its debt?*
 - Inability to refinance
 - People lose confidence and stop lending
 - Interest rates increase
- *How does growth in GDP affect a country's ability to pay debt?*
 - Ability to pay grows as the economy grows (larger tax base)
 - Other countries can demand dollars at a faster rate than US GDP growth
- *What can we associate with the reasons that dominant currencies fall?*
 - The Triffin dilemma
- *Will the dollar remain the dominant currency in 5 years? 10 years? 200 years?*
 - Likely in the short term but not the long term
 - Dollar alternatives and poor US fiscal health could erode confidence in the dollar
 - Timing is uncertain

Segment 3: Tariff policy and the US dollar (15 min.)

The instructor can now turn the discussion to a recent shock that affected the value of the dollar. To start this segment, the instructor can ask, “When US tariffs were instituted in 2025, what happened to the dollar?” Students should know that the dollar generally depreciated (case Exhibit 5a) following the initial tariff announcement in April. If the instructor taught the companion case, students can be reminded how uncertainty impacts the value of the dollar, and then the instructor can ask, “Aside from uncertainty, what are the possible effects of tariffs?” Students should recognize that people might sell US assets (which would weaken the dollar), leading to a rise in net capital outflow (NCO) and higher interest rates. The instructor can diagram these impacts on the board using the Three-Paned Model (see **Board plan 2**). For this scenario, the instructor can note that net exports and output grow (an intended effect of the 2025 US tariffs) but focus on the impact on interest rates. To relate the effects to the previous discussion, the instructor can ask, “What happens when interest rates increase?”⁷ Student should remember that higher interest rates can affect the government’s ability to repay debt; if people lose their appetite for dollar-denominated assets and interest rates soar, the US government could potentially default on its debt. At this point, the instructor could mention that during global crises or times of uncertainty, capital generally tended to flow into the United States as investors sought safety, which kept interest rates relatively low. (The instructor can also emphasize this point in Segment 4, when discussing interest rates and debt sustainability.)

The following questions can facilitate discussion for this segment:

- *When US tariffs were instituted in 2025, what happened to the dollar?*
 - It was depreciated
- *Aside from uncertainty, what are the possible effects of tariffs?*
 - The selling of US assets

⁷ Case Exhibit 5d shows interest rate data for 10-Year US Treasury securities.

- An increase in NCO
- An increase in interest rates
- *What happens when interest rates increase?*
 - A decrease in the ability to service government debt
 - Potential default

Segment 4: Debt sustainability (20 min.)

After discussing some of the issues facing the United States, the instructor can return to the central topic of debt sustainability by linking the concept to the government budget identity in case Appendix 2 and **Board plan 3**. The formula implies that the government must run future budget surpluses to back the current debt. Put another way, the current government debt must be less than or equal to the present discounted value of all primary fiscal balances.

To facilitate discussion during this portion of the segment, the instructor could ask, “What factors influence the government debt?” Here, students should just recognize that the government debt accumulates when government spending exceeds tax revenue (to meet its obligations, the government must either collect funds through taxation or borrowing). The instructor can begin writing the budget formula from **Board plan 3** and ask, “Would someone be willing to make a loan if the present value of future income is not enough to pay for the loan?” Students should indicate that people might become less willing to make a loan if they lose belief in the borrower’s ability to pay back the debt, which could cause interest rates to increase. To further the discussion, the instructor can ask, “According to the formula, what happens when interest rates increase?” Looking at the formula, students should see that higher interest rates would lower the net present value of the future budget surplus, implying that debt repayment would become more difficult. The instructor could emphasize that, once interest rates reach a certain level, people will stop lending and a dominant currency can suddenly lose its status (“Gradually, then suddenly.”⁸). Conversely, when investors deem an interest-paying asset as safe, they increase their demand for the asset during times of uncertainty, which can help keep interest rates relatively low. In this segment, the instructor could also use the formula to stress that increasing levels of government debt would require higher tax revenues or reduced government spending to service the obligations over time, making repayment more challenging—or the instructor could make this point during the ensuing discussion about US government debt.

The instructor could reference the US debt-sustainability projections in case Exhibits 7a and 7b before turning to specific options for lowering the US government debt burden. The instructor can ask, “How can the United States decrease its debt burden?” and review the likelihood of each of the alternatives from case Appendix 2 (also see **Board plan 3**). After initial dialogue, the instructor could ask, “What is the most viable option?” Answers might vary, but the instructor can choose to focus discussion on the option of raising taxes (specifically, a value-added [consumption] tax [VAT]), which some people deem the best alternative due to the low rates of taxation in the United States compared to other countries. Contrarily, some students might point out that a value-added tax disproportionately hurts those who need to spend money (the less wealthy) or that higher taxes are not politically popular. At this point, the instructor could ask students for other options or could reference the idea of investor Warren Buffett, who suggested that members of Congress could become ineligible for reelection if the annual budget deficit exceeds 3% of GDP.⁹

⁸ Ernest Hemingway, *The Sun Also Rises* (Scribner, 2006).

⁹ Alex Crippen, “Warren Buffett’s 5-Minute Plan to Fix the Deficit,” CNBC, July 7, 2011, <https://www.cnbc.com/2011/07/07/warren-buffetts-5minute-plan-to-fix-the-deficit.html#:~:text=UPDATE:%20An%20attorney%20in%20St,take%20more%20than%20five%20minutes> (accessed Jan. 22, 2026).

The following questions can facilitate discussion for this segment:

- *What factors influence government debt?*
 - Government debt accumulates when government spending exceeds tax revenue
- *Would someone be willing to make a loan if the present value of future income is not enough to pay for the loan?*
 - People would be less willing to loan money
 - If people lose confidence in the debtor's ability to repay, interest rates can increase
- *According to the formula, what happens when interest rates increase?*
 - The net present value of future surpluses (the ability to repay) decreases
- *How can the United States decrease its debt burden?*
 - See case Appendix 2 and **Board plan 3**
- *What is the most viable option?*
 - Possibly a VAT
 - But a VAT disproportionately hurts some and might not be politically feasible

Segment 5: Firm decision (10 min.)

Following the discussion about the threats to the dollar's dominance, the instructor can return to Marfrig's decision, opening this segment by asking a question relevant to the firm's choice: "Is the recent dollar decline a temporary or structural shift in the preference for dollar-denominated assets?" Some students might argue that uncertainty related to US tariff policy or fiscal sustainability have caused faith in US institutions to begin to erode, signaling a structural change. These students might point to the rise in bitcoin and gold prices as signs that viable dollar alternatives exist. Other students might question the credibility of potential dollar substitutes and the eventual fall of the dollar, arguing that the dollar remains relatively well positioned, and that recent dollar weakness simply represents a short-term reaction to policy changes. Depending on the nature of the previous discussion, if the instructor still needs to connect the policy environment to the firm's decision, they could ask, "How does policy affect the short-term and long-term outlook for the dollar?"

After students consider the current and future issues that affect the dynamics of the dollar, the instructor can ask, "How should Marfrig anchor its new contracts?" Students will offer different answers, but responses should generally connect to the dialogue from the beginning of this segment, with students reflecting upon policy implications and the time horizons for the business deals. For example, students concerned about the direction of US policy and the long-term prospects of the dollar might prefer to complete the 30-year bond issuance with a different currency.¹⁰ Since student opinions should vary, the instructor can conclude by relating the difficulty of the firm's decision to the uncertain timing of the fall of dominant currencies.

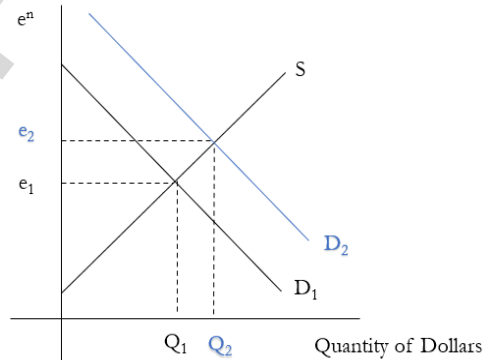
The following questions can facilitate discussion for this segment:

- *Is the recent dollar decline a temporary or structural shift in the preference for dollar-denominated assets?*
- *How does policy affect the short-term and long-term outlook for the dollar?*
- *How should Marfrig anchor its new contracts?*

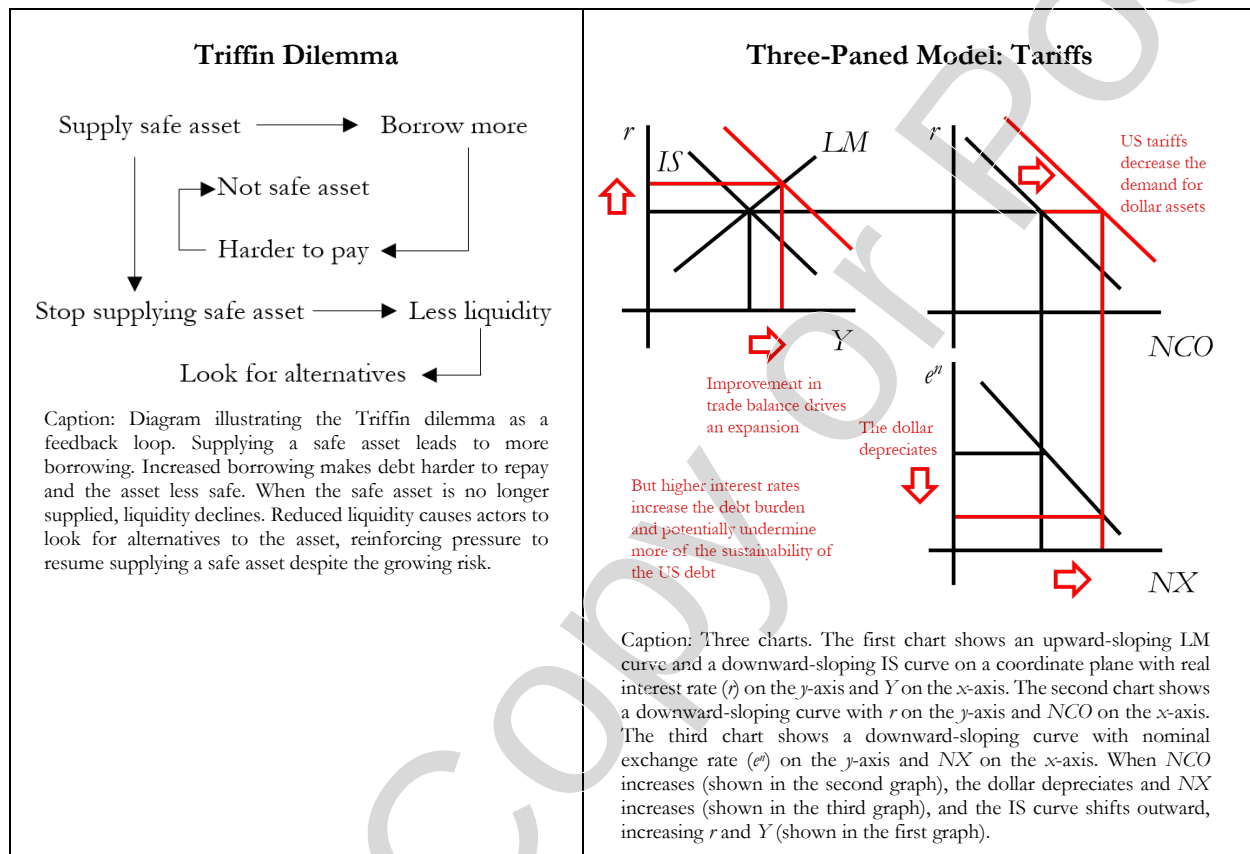
¹⁰ Although not central to the case, students could reference exchange-rate trends for Brazil (case Exhibit 9) as relevant to the decision.

Board Plans

Board plan 1. The US dollar and dollar-denominated assets.

The US Dollar: Dominant Currency	Supply and Demand for US Dollar-Denominated Assets
Firms invoice and borrow in dollars → “natural” hedge • Network externalities/coordination • Safety • Stability • Liquidity • Sound institutions	Factors affecting supply • US government debt • Federal Reserve policy • US demand for imports • US investment in foreign countries Factors affecting demand • Incomes in foreign countries • Foreign investment in the United States • Foreign demand for US exports High demand → the dollar appreciates  Caption: This chart shows an upward-sloping supply curve (S) and a downward-sloping demand curve (D_1 and D_2) on a coordinate plane with the exchange rate (e^n) on the y-axis and quantity of dollars (Q_1 and Q_2) on the x-axis. When the demand curve shifts outward, the dollar appreciates.

Board plan 2. The Triffin dilemma and US tariff policy.

Board plan 3. US debt sustainability.¹¹

Debt Sustainability	US Debt-Sustainability Options
$B_0 \leq \sum_{t=1}^{\infty} \frac{T_t - G_t}{(1+r)^n}$	• Grow very fast • Default • Print money • Cut spending • Raise taxes*

* Mankiw viewed this as the most likely course of action.

¹¹ B_0 = current government debt; T_t = future tax revenue at time t ; G_t = future government expenditures at time t ; r = discount rate; n = number of periods.